[TODO: Title after the holdout evaluation]

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October 15, 2026

Abstract

[TODO: abstract: written after the holdout evaluation.]

Keywords: Expected Shortfall, Value-at-Risk, volatility spillovers, contagion, deep learning, elicitability, pre-registration.
JEL: C45, C52, C53, C58, G15, G17.

1 Introduction

[TODO: intro: written after the holdout evaluation.]

2 Information set: what is known before a market opens

Equity markets in different time zones trade in sequence. When the risk manager of market i sets the Value-at-Risk for trading day D shortly before i opens, markets that have traded in the meantime have already revealed information that i's own close-to-close history does not contain (Hamao et al., 1990; Lin et al., 1994). Let $C_i(D)$ and $O_i(D)$ denote the close and open of market i on its local trading day D, in coordinated universal time, and let D^- be i's previous trading day.

Definition 1 (Pre-open information set)**.** *$\mathcal{F}_{i,D}^{\text{pre}}$ contains the returns of market i up to D^- and, for every other market j, all returns of sessions s with $C_j(s) \le O_i(D) - 2m$, where m is a safety margin.*

The new information for day D is the overnight move of each foreign market,

$$x_{j,D} = \sum_{s:\, C_i(D^-) < C_j(s) \le O_i(D) - 2m} \frac{r_{j,s}}{s_{j,s}}, \tag{1}$$

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where $s_{j,s}$ is j's trailing 250-day root mean square return up to the previous session, so that $x_{j,D}$ is measured in units of j's usual volatility. An indicator $a_{j,D}$ records whether at least one such session exists.

Historical exchange schedules have changed (Tokyo extended its close by 30 minutes in 2024, Seoul in 2016, Brazil abolished daylight saving in 2019). To rule out look-ahead bias, we use the *latest* plausible close of every exchange, including closing auctions, and the *earliest* plausible open, including pre-opening sessions. We then require an additional margin of $m = 30$ minutes on each side, and handle daylight saving with the IANA time-zone database. A violation could only arise from a schedule error of more than an hour in the conservative direction. The construction makes stale information look new, which can only weaken the predictive content of the news, never strengthen it. In our development sample, Asian markets see the previous U.S. and European sessions on about 97% of days, and U.S. markets see the same day's Asian sessions on about 95% of days.

3 Scoring and statistical efficiency

As in Agrawal (2026), forecasts of $(\mathrm{VaR}_\alpha, \mathrm{ES}_\alpha)$ for $\alpha \in \{1\%, 2.5\%, 5\%\}$ are trained and evaluated with the strictly consistent FZ0 loss of Patton et al. (2019),

$$L_\alpha(y, v, e) = -\frac{1}{\alpha e}\mathbf{1}\{y \le v\}(v - y) + \frac{v}{e} + \log(-e) - 1.$$

FZ0 receives information about its target mainly from tail days. Away from the tail, the loss is $v/e + \log(-e) - 1$, which depends on the forecast only through its level. Learning *dynamics*, i.e. how risk responds to news, from FZ0 alone is therefore statistically inefficient. A Gaussian quasi-likelihood instead learns the dynamics of volatility from every day's squared return. This motivates an auxiliary score.

Proposition 1 (Composite score)**.** *Let $Q(y, \varsigma^2) = y^2/\varsigma^2 + \log \varsigma^2$ (QLIKE). For $\lambda \ge 0$, the score*

$$S(y; \{v_k, e_k\}_{k=1}^K, \varsigma^2) = \sum_{k=1}^K L_{\alpha_k}(y, v_k, e_k) + \lambda\, Q(y, \varsigma^2)$$

is strictly consistent for the vector $(\{\mathrm{VaR}_{\alpha_k}, \mathrm{ES}_{\alpha_k}\}_{k=1}^K,\ \mathbb{E}[y^2])$, provided $\mathbb{E}y^2 < \infty$ and every L_{α_k} is strictly consistent for its pair (Proposition 1 of Agrawal, 2026).

Proof. $\mathbb{E}Q(Y, \varsigma^2) = \mathbb{E}[Y^2]/\varsigma^2 + \log \varsigma^2$ has derivative $(\varsigma^2 - \mathbb{E}Y^2)/\varsigma^4$ and is uniquely minimised at $\varsigma^2 = \mathbb{E}Y^2$. The expected composite score is a sum of terms that depend on disjoint components of the report, each uniquely minimised at the corresponding true functional, so the sum is uniquely minimised at the true vector. □

Remark 1. In a location-scale network $v_k = \varsigma q_k$, $e_k = \varsigma c_k$, the reports are linked. The composite score then still identifies the truth when the model is correctly specified. Under misspecification, λ

trades off tail fit against volatility fit. We fix $\lambda = 1$ a priori and do not tune it. QLIKE is also the loss that is robust to noise in the volatility proxy (Patton, 2011).

4 Models

All models produce one-step-ahead (VaR, ES) forecasts at the three levels. The benchmarks of Agrawal (2026) (historical simulation, GARCH-type models with normal, Student-t and skewed-t innovations, filtered historical simulation, GARCH-EVT, CAViaR-ES and two FZ-GAS models) and its deep models (LSTM-FZ, TRF-FZ, GJR-LSTM) use only the close-to-close information set and are carried over unchanged. Four models are new.

GJRX-skt (econometric, pre-open information). A GJR-GARCH(1,1,1) with Hansen skewed-t innovations whose variance responds linearly to squared foreign news,

$$\sigma_t^2 = \omega + (a + g\,\mathbf{1}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + b\,\sigma_{t-1}^2 + d_1\bar{X}_t + d_2\bar{X}_t^-, \tag{2}$$

where $\bar{X}_t$ is the average of $x_{j,t}^2$ over markets with fresh news and $\bar{X}_t^-$ the same average restricted to negative moves. Both are known before r_t is realised. The model is estimated by maximum likelihood with $d_1, d_2 \geq 0$ on the same rolling window and refit schedule as the other econometric models. It nests GJR-skt ($d_1 = d_2 = 0$).

NG-FZ (neural news-impact GARCH). Returns are scaled by a per-asset constant, $y_t = r_t/s$, and a d-dimensional state follows a *linear* recursion driven by a *learned* news-impact function (Engle and Ng, 1993):

$$\begin{aligned} h_t &= A\,h_{t-1} + N_\theta(y_{t-1}), \qquad A = \mathrm{diag}(a_1, \ldots, a_d), \quad 0 < a_k < \rho < 1, \\ N_\theta(y) &= B\,(y,\, y^2,\, y^2\mathbf{1}\{y<0\},\, |y|)' + c + g \cdot \mathrm{MLP}(\tanh(y/2), \log(1+y^2), \mathbf{1}\{y<0\}). \end{aligned} \tag{3}$$

The first coordinate is a GJR-GARCH variance, the others are volatility memories at decay rates $0.5, \ldots, 0.995$ (as in component GARCH). Log-variance and tail shape are read out by small networks, and the output uses the monotone head of Agrawal (2026).

Proposition 2 (Stability and nesting)**.** *(i) For any input path and any two initial states, $\|h_t - h_t'\|_\infty \leq \rho^t \|h_0 - h_0'\|_\infty$: the filter forgets its initialisation exponentially (invertibility). (ii) If (y_t) is strictly stationary and ergodic with $\mathbb{E}\log^+\|N_\theta(y_0)\| < \infty$, then $h_t^* = \sum_{j\geq 0} A^j N_\theta(y_{t-1-j})$ converges almost surely, is the unique stationary ergodic solution of* (3)*, and $\|h_t - h_t^*\| \to 0$ exponentially almost surely (Brandt, 1986; Bougerol, 1993). (iii) With $g = g' = 0$ (both neural gates closed) and $d = 1$, NG-FZ is exactly GJR-GARCH(1,1) with a constant standardised tail.*

Proof. (i) $h_t - h_t' = A^t(h_0 - h_0')$ because the input enters additively and does not depend on the state; $\|A^t\|_\infty = \max_k a_k^t \leq \rho^t$. (ii) By the root test, $\limsup_j \|A^j N_\theta(y_{t-1-j})\|^{1/j} \leq \rho \cdot \limsup_j \exp(\log^+\|N_\theta(y_{t-1-j})\|/j) =$

$\rho < 1$ almost surely, because $\mathbb{E}\log^+\|N_\theta(y_0)\| < \infty$ implies $\log^+\|N_\theta(y_{-j})\|/j \to 0$ (Borel–Cantelli). Uniqueness and the convergence of h_t follow from (i). (iii) By substitution. □

The network is initialised at (iii), with the GARCH part at a fitted GJR model and the tail at the empirical tail of its residuals, so that training starts from a strong econometric model and learns corrections by minimising FZ0. Gradient descent on a linear recursion with bounded decays cannot produce explosive filters, which is a practical advantage over generic recurrent networks.

CTG-FZ (contagion network). CTG-FZ adds a cross-market module to LSTM-FZ, with identical inputs and hyperparameters otherwise. Each foreign market contributes a token $k_j = \mathrm{MLP}(x_j, |x_j|, \mathrm{emb}_j)$. A query is formed from the target market's embedding and its LSTM state, and attention over the markets with fresh news yields a contagion vector $c = \sum_j w_j k_j$, which is zero when no foreign market traded overnight. Because the tokens are non-linear in x_j and the query depends on the target's own volatility state, the module can represent disproportionate responses to large foreign crashes, market-specific lead–lag structure and state-dependent contagion. The weights w_j form an interpretable transmission matrix. In the final specification (v2, Section 5) the output is location–scale, $\mathrm{VaR} = s \varsigma q$ and $\mathrm{ES} = s \varsigma c$, with ς from a variance head trained by the composite score of Proposition 1.

LSTMQ-FZ (ablation). LSTMQ-FZ is identical to CTG-FZ v2 with the foreign news switched off. The comparison CTG-FZ versus LSTMQ-FZ therefore isolates the value of the cross-market information for the network, holding architecture and objective fixed.

Combinations. COMB-All (all econometric models except HS, plus the three deep models of Agrawal, 2026) is extended to COMB-All+X by adding GJRX-skt and CTG-FZ, and to COMB-All+NG by adding NG-FZ. All combinations are equal-weight averages, specified before any result was seen.

5 Pre-registered design

Samples. The *development* sample consists of the twelve indices of Agrawal (2026): S&P 500, Nasdaq Composite, Russell 2000, FTSE 100, DAX, CAC 40, Nikkei 225, Hang Seng, BSE Sensex, S&P/ASX 200, KOSPI and Bovespa. The *holdout* sample consists of twelve further indices: S&P/TSX, SMI, TAIEX, Shanghai Composite, IPC, AEX, IBEX 35, KLCI, Jakarta Composite, Straits Times, ATX and BEL 20. The holdout markets were chosen by a mechanical rule, before any model was run on them. The rule requires at least 2,000 daily observations before 2006, no multi-month gaps and fewer than fifteen stale prices; it excluded OMXS30, Merval and FTSE MIB. Both samples use the out-of-sample period January 2006 to September 2026, a 2,000-day rolling window and the evaluation methods of Agrawal (2026).

Protocol. The protocol, its amendments and every development result are recorded in a public, time-stamped file (`docs/PROTOCOL.md` in the repository). New models were designed and revised on the development sample only. Before any holdout run, the final specifications were frozen in a git commit (`7a58b26`, 15 October 2026), and the holdout sample was then evaluated exactly once. The pre-registered hypotheses, at $\alpha = 2.5\%$ with panel Diebold–Mariano tests, were:

- H1 CTG-FZ has lower FZ0 loss than GJRX-skt (learning contagion versus a linear spillover model with the same information);
- H2 CTG-FZ has lower loss than LSTM-FZ (value of the cross-market information for the network);
- H3 COMB-All+X has lower loss than COMB-All.

Development history. For transparency we report every variant tried on the development sample (Table 1). NG-FZ v1 shared one GARCH block across markets. v2 gave each market its own block and overfitted. The diagnosis, that FZ0 learns volatility dynamics inefficiently, motivated v3 (v2 plus the auxiliary QLIKE score of Proposition 1), which roughly halved v2's deficit. Under the pre-specified rule (lowest development loss), v1 was frozen. CTG-FZ v1 reacted to foreign news but did not improve on LSTM-FZ. By the same efficiency diagnosis, v2 adopted the location–scale output with the composite score, and the ablation LSTMQ-FZ was added at that point. A bug in the first v2 run (a variable that silently disabled the new settings) was detected before any result was seen, fixed with a test, and the run was repeated. v2 was frozen by the selection rule.

Table 1: Development history (development sample, $\alpha = 2.5\%$): FZ0 loss relative to the stated comparison

Variant	Comparison	Loss difference	DM t
NG-FZ v1 (pooled GARCH block)	GJR-skt	+0.0044	0.42
NG-FZ v2 (market-specific block)	GJR-skt	+0.0272	2.18
NG-FZ v3 (v2 + QLIKE)	GJR-skt	+0.0146	1.65
CTG-FZ v1	GJRX-skt / LSTM-FZ	+0.0177 / +0.0016	1.30 / 0.96
CTG-FZ v2	GJRX-skt / LSTM-FZ	+0.0153 / −0.0007	1.22 / −0.29
CTG-FZ v2	LSTMQ-FZ (ablation)	+0.0002	0.48
GJRX-skt	GJR-skt	−0.0193	−3.90

Frozen for the holdout: NG-FZ v1, CTG-FZ v2, LSTMQ-FZ, GJRX-skt and all models and combinations of Agrawal (2026).

6 Results

[TODO: results: written after the holdout evaluation.]

7 Conclusion

[TODO: conclusion: written after the holdout evaluation.]

Acknowledgments and disclosure

The author used AI-based assistance (large language model tools) for code development, literature search and editing. All modelling choices, results and interpretations were checked by the author, who takes full responsibility for the content of this paper. Any errors are the author's own.

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